

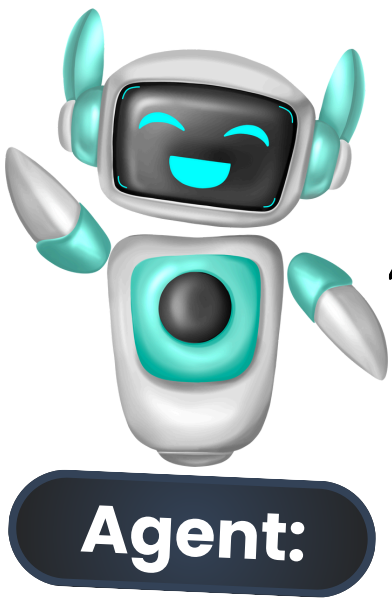
Greg Coquillo
Product Leader

HOW TO BUILD AI AGENTS FROM SCRATCH



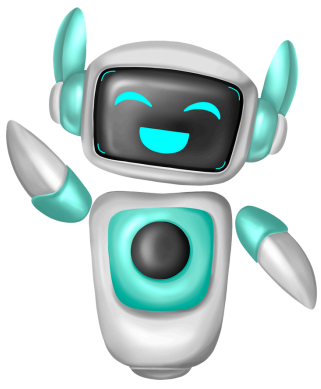
Greg Coquillo
Product Leader

DEFINE THE AGENT'S ROLE GOAL





What specific task will your agent perform, and what outcome should it deliver?



Who is it helping?

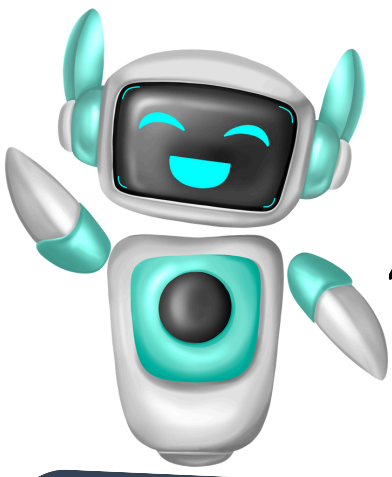
Identify the user or team it will assist.



A Logistics Assistant that analyzes warehouse data, predicts inventory shortages, and recommends restocking dates.

Greg Coquillo
Product Leader

DESIGN STRUCTURED INPUT & OUTPUT

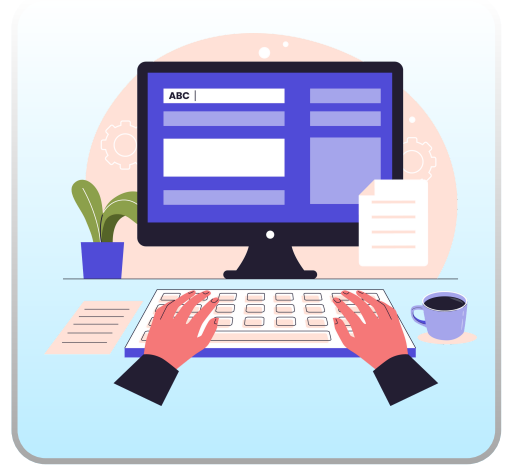


Agent:



User:

Input



Use JSON, forms, or structured prompts so the agent always receives information in a predictable format.

Output



Decide how the response should be formatted - text summaries, tables, charts, or JSON.

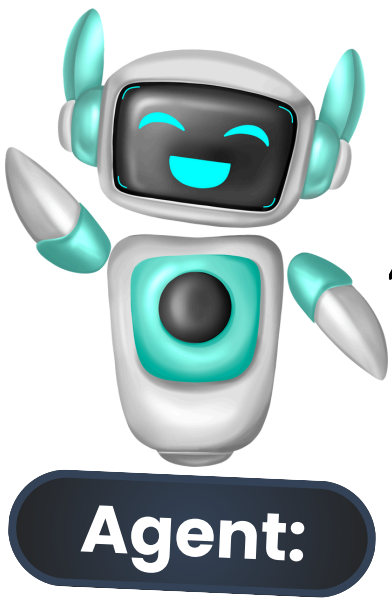


Avoid messy text - think like like an API

Tool: LlamaParse, LangChain Output Parsers

Greg Coquillo
Product Leader

PROMPT AND TUNE THE AGENT'S BEHAVIOUR



Agent:

"You are a cybersecurity advisor who gives concise risk assessments and recommends mitigation steps."

Prompt

Prefix Tuning



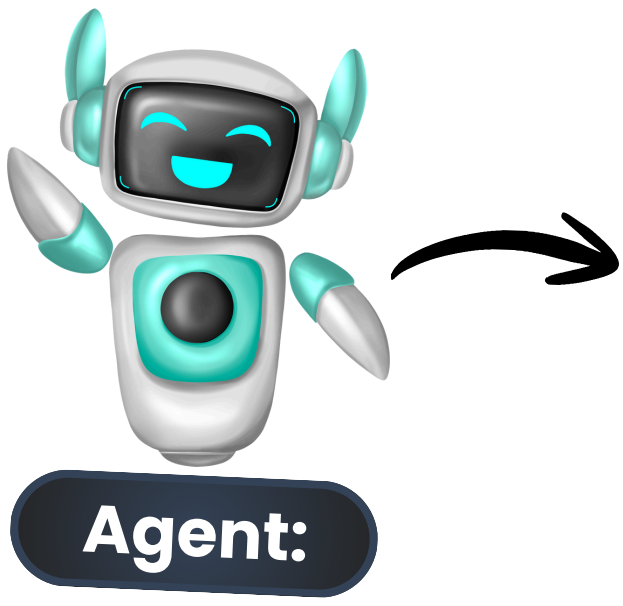
User:

- Start with role-based systems prompt
- Use Prompt Turning or prefix Tuning for consistent persona and task behaviour

Tool: GPT-4, Claude Tuning, PromptScript

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Product Leader

ADD REASONING AND TOOL USE

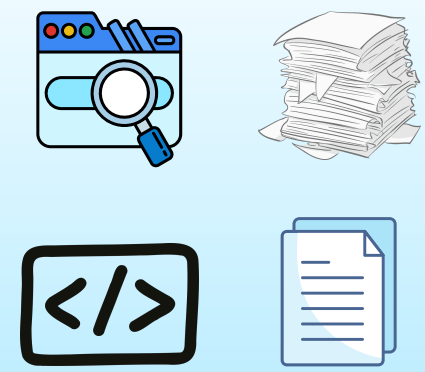


Agent:

Reasoning Frameworks

- ReAct (Reasoning + Action)
- Chain-of Thought

Tools





User:

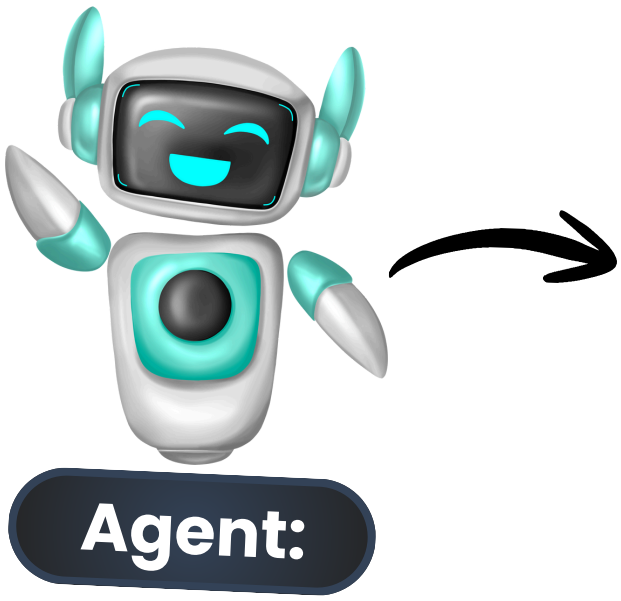
- Equip wihe agent with reasoning frameworks
- Allow access the tools like web search, code interpreters. or document retireverse.

Tool: LangChain,OPEAI Tools,ReAct Frameworks

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Product Leader

STRUCTURE MULTI-AGENT LOGIC

Define specialized agents and how they collaborate.





Planner Agent

Creates a workflow

Analyst Agent

Processes data

Writer Agent

Generates documents

Reviewer Agent

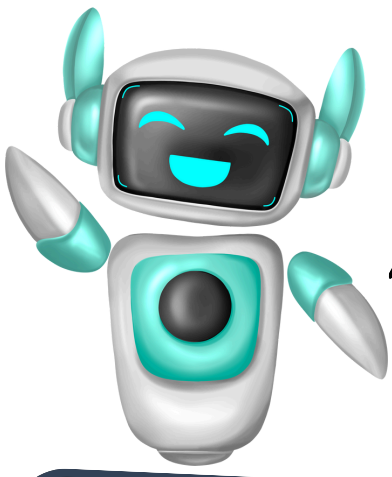
Validates accuracy

- Use archestration frameworks to define agent roles.
- Create Planner, Researcher, Report agents - each with oens interpreters, document input-./xpout schema.

Tool: CrewAI, LangGraph, OPENai Swarm

Greg Coquillo
Product Leader

GIVE YOUR AGENT MEMORY & CONTEXT CONTINUITY



Agent:

MEMORY

- Vector-based memory
- Summary memory

RAG

(Retrieval- augemnted Generation)

Knowledge Base



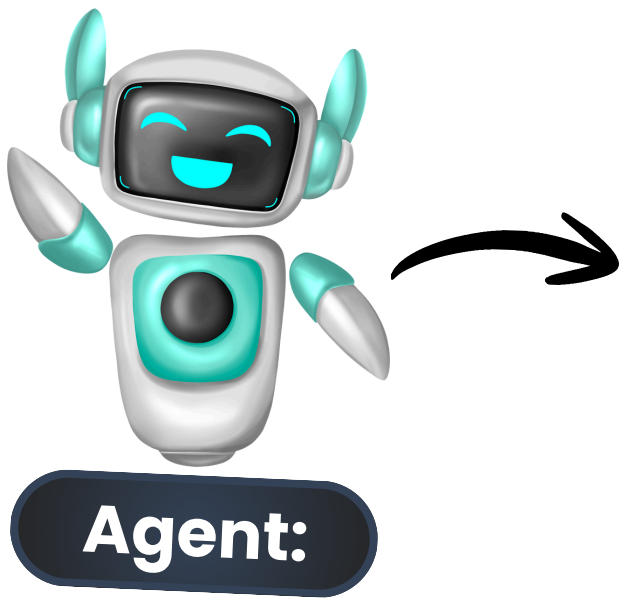
User:

Does the agent need to recall previous interactions, tasks, or documents?

Tool: ChromaDB, FAISS, Zep, LangChain Memory


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Product Leader

ADD VOICE OR VISION CAPABILITIES



Agent:

Agent Capabilities



Voice

Let the agent listen and speak naturally.

Coqui, ElevenLab

Vision



Allow the agent to interpret images, charts, screenshots, or documents.



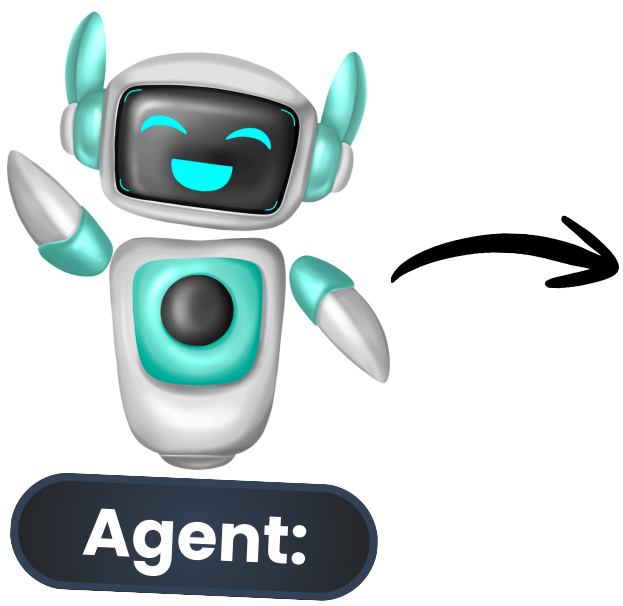
User:

- Let your agent see and speak, memory
- Use conversational memory, vector-based memory.

Tool: Zep, LangChaibs,GPT-O, ChromaD 2 FAISS

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DELIVER THE OUTPUT



Output Formating



Markdown → JSON



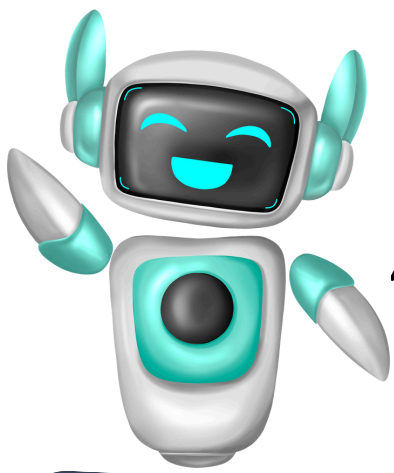
- Format outputs into MarkDown → PDF or JSON
- Output must be readable and parsoble.

Tool: Pydantic AI, LangChain Parser



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Product Leader

WRAP IN A UI



Agent:

Output Formating



Markdown → JSON



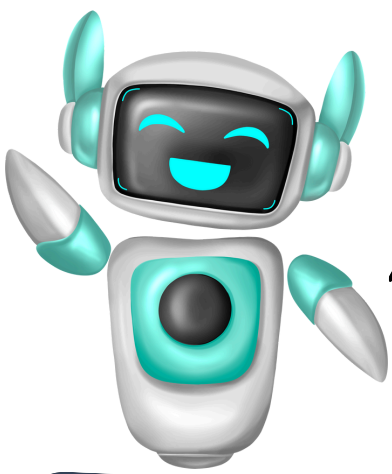
User:

- Create a front-end
- This is what turns your agent into a product

Tool: Gradio,Streamlit

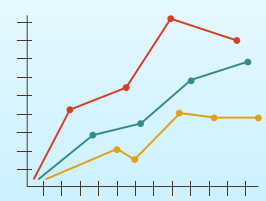
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Product Leader

EVALUATE AND MONITOR



Agent:

Monitoring & Evaluation



Logs





Feedback

test prompts



User:

- Run test prompts and toolchains to check reliability.
- Use logs, benchmarks and feeback to improve over time.

Tool: MCP Logs, OPENAI Evaluation API, Custom Metrics Dashbaords.



Greg Coquillo

Product Leader